

3.2 Kinematics in 1 Dimension

Mathematics B (MEI) — OCR B — A-Level (H640) — Mechanics — spec refs k1–k8

What this topic covers. The precise vocabulary of motion in a straight line, motion graphs and what their gradients and areas mean, the constant acceleration formulae, and the use of calculus when acceleration is not constant.

1. The vocabulary — and why it matters

MEI distinguishes several pairs that are casually treated as synonyms (spec refs k1–k3).

Scalar	Vector	Distinction
Distance travelled	Displacement	Total path length vs net change of position, with direction
Speed	Velocity	Magnitude only vs magnitude and direction
Magnitude of acceleration	Acceleration	Never negative vs signed

Position is measured from a fixed origin; **displacement** is the change in position from a starting point.

Worked example. A body travels 100 m north then 100 m south in 40 s.

$$\text{average speed} = \frac{200}{40} = 5 \text{ m s}^{-1}, \quad \text{average velocity} = \frac{0}{40} = 0$$

$$\text{average speed} = \frac{\text{distance travelled}}{\text{elapsed time}}, \quad \text{average velocity} = \frac{\text{overall displacement}}{\text{elapsed time}}$$

2. Motion graphs

Graph	Gradient gives	Area underneath gives
Displacement-time	Velocity	No meaning
Velocity-time	Acceleration	Displacement
Acceleration-time	—	Change in velocity

- A straight line on a displacement-time graph means constant velocity.
- A horizontal line on a velocity-time graph means constant velocity; a straight sloping line means constant acceleration.
- Area **below** the time axis on a velocity-time graph is displacement in the negative direction.

Worked example. A trapezium on a velocity-time graph has parallel sides 4 and 10 and width 6. The displacement is

$$\frac{1}{2}(4 + 10) \times 6 = 42 \text{ m}$$

Worked example. A body travels at 12 m s^{-1} for 5 s, then decelerates uniformly to rest over the next 4 s. Find the total distance.

Rectangle: $12 \times 5 = 60$. Triangle: $\frac{1}{2} \times 12 \times 4 = 24$. Total 84 m.

For **distance travelled** rather than displacement, treat any area below the axis as positive. Adding signed areas gives displacement; adding magnitudes gives distance.

3. The constant acceleration formulae

With u initial velocity, v final velocity, a acceleration, s displacement and t time:

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$s = vt - \frac{1}{2}at^2$$

$$s = \frac{1}{2}(u + v)t$$

$$v^2 = u^2 + 2as$$

Each omits one variable, so choose the one that avoids the quantity you neither know nor want. The fifth omits t ; the first omits s .

These apply **only** when the acceleration is constant. If acceleration is given as a function of time, they are invalid and calculus is required.

Worked example. A particle starts from rest and accelerates at 2 m s^{-2} for 5 s.

$$v = 0 + 2(5) = 10 \text{ m s}^{-1}, \quad s = 0 + \frac{1}{2}(2)(25) = 25 \text{ m}$$

Worked example. A car travelling at 20 m s^{-1} decelerates uniformly at 4 m s^{-2} to rest.

$$0 = 20 - 4t \implies t = 5 \text{ s}, \quad s = \frac{1}{2}(20 + 0)(5) = 50 \text{ m}$$

Worked example. A body has $u = 5 \text{ m s}^{-1}$ and accelerates at 3 m s^{-2} over 10 m.

$$v^2 = 25 + 2(3)(10) = 85 \implies v = 9.22 \text{ m s}^{-1} \text{ (3 s.f.)}$$

Vertical motion under gravity

Taking $g = 9.8$ and being consistent about the positive direction:

Worked example. A stone is dropped from rest. After 2 s,

$$v = 9.8(2) = 19.6 \text{ m s}^{-1}, \quad s = \frac{1}{2}(9.8)(4) = 19.6 \text{ m}$$

Worked example. A stone falls from rest through 45 m. Find the time.

$$45 = \frac{1}{2}(9.8)t^2 \implies t^2 = 9.184 \implies t = 3.03 \text{ s (3 s.f.)}$$

Worked example. A ball is thrown vertically upwards at 14.7 m s^{-1} . Find the maximum height. At the highest point $v = 0$:

$$0 = 14.7^2 - 2(9.8)s \implies s = \frac{216.09}{19.6} = 11.025 \text{ m}$$

Fix the positive direction at the start and keep it. If upwards is positive, then $a = -9.8$ throughout — including on the way up, at the top, and on the way down. The acceleration never changes; only the velocity does.

4. Calculus for variable acceleration

When acceleration is not constant, differentiate to go down the chain and integrate to go up (k5):

$$v = \frac{ds}{dt}, \quad a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$$

$$v = \int a \, dt, \quad s = \int v \, dt$$

Worked example. For $s = t^3 - 3t$, find the velocity at $t = 2$.

$$v = 3t^2 - 3 \implies v(2) = 12 - 3 = 9 \text{ m s}^{-1}$$

Worked example. A particle has $a = 6t$ and velocity 2 m s^{-1} at $t = 0$.

$$v = \int 6t \, dt = 3t^2 + c, \quad v(0) = 2 \Rightarrow c = 2$$

$$v = 3t^2 + 2$$

The constant of integration is fixed by the initial condition. Omitting it, or forgetting to apply the condition, is the standard error in these questions.

Distance travelled when the direction reverses

Worked example. For $s = t^3 - 6t^2 + 9t$, find the times at rest and the total distance travelled from $t = 0$ to $t = 4$.

$$v = 3t^2 - 12t + 9 = 3(t - 1)(t - 3) = 0 \implies t = 1 \text{ and } t = 3$$

Evaluating the displacement at the key times:

$$s(0) = 0, \quad s(1) = 4, \quad s(3) = 0, \quad s(4) = 4$$

The particle moves forward 4 m, back 4 m, then forward 4 m again. So the **distance travelled** is 12 m, while the **displacement** is only 4 m.

Whenever the velocity changes sign inside the interval, split at those instants and add the magnitudes. Integrating straight through gives the displacement, not the distance — and the question usually wants the distance.

5. Relative velocity in one dimension

The velocity of A relative to B is $v_A - v_B$. Two cars travelling the same way at 30 and 20 m s^{-1} have a relative velocity of 10 m s^{-1} — which is why the gap closes slowly despite both moving quickly.

6. Common pitfalls

- Using the suvat formulae when the acceleration is not constant.
- Changing the sign of g partway through a vertical motion problem.
- Giving displacement when the question asks for distance travelled.
- Omitting the constant of integration, or not applying the initial condition.
- Confusing the gradient and the area of a velocity-time graph.
- Assuming acceleration is zero at the highest point of a vertical throw — the velocity is zero, not the acceleration.
- Reporting a negative distance.